SYDNEY
BOYS
HIGH
SCHOOL

2019

YEAR 9 YEARLY MATHEMATICS

General Instructions:

- All questions may be attempted.
- Write using black or blue pen.
- Marks may be deducted for careless or badly arranged work
  This can include insufficient working out.
- All working and answers are to be written in this test booklet.
- If you wish to rewrite an answer, draw a line through your faulty answer and rewrite your answer on pages 26 and 27 of this booklet. Show the number and part of the answer being rewritten and write "See Back" at the original question.
- NESA approved calculators may be used.
- Clearly indicate your class by placing an **X** next to your class.

Time Allowed: 90 minutes

Examiner: AW

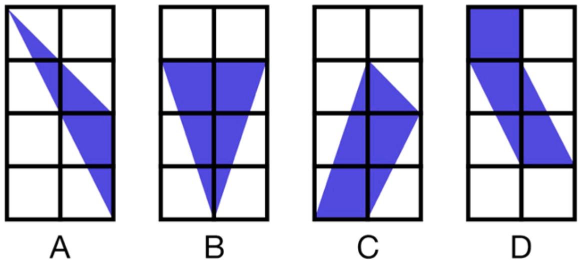
Name: ____

Class	Teacher	
9MaA	Mr A Wang	
9MaB	Mr James	
9MaC	Ms H Chan	
9MaP	Mr Fuller	
9MaL	Ms J Chan	
9MaU	Ms Kilmore	
9MaS	Ms Millar/Mr R Wang	

Question	Marks
1	/ 16
2	/ 17
3	/ 16
4	/ 16
5	/ 17
6	/ 18
Total	/ 100

Question 1 (16 marks)

a) Circle the letter of the polygon that does not have the same area as the others. **1**



b) Expand then collect like terms for the following expressions:

(i) $4(a+1)-6a-5$ **1**

(ii) $(2a+b)(a-11b)$ **1**

c) JoJay, a casual labourer, is paid $21.78 per hour. How much will he earn if he works a 35 hour week? **1**

d) Show by substitution that $a = 2$ is the solution to the following equation. **1**

$11(a + 8) = 5(a + 20)$

e) Write $2500 \times 3\ 640\ 000$ in scientific notation. **1**

Question 1 Continued

f) Solve the following equations, giving answers in simplest exact form.

(i) $4p+6=37$ **1**

(ii) $\frac{y}{5}+3=7$ **1**

(iii) $2y^2+9=23$ **2**

(iv) $\frac{2x-1}{4}=\frac{3x+2}{8}$ **2**

Question 1 Continued

g) Express $5x^{-3}$ with a positive index. **1**

h) Find the gradient *m* and *y*-intercept of these lines:

(i) $y = 3x + 2$ **1**

(ii) $4x + 3y + 6 = 0$ **2**

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Question 2 (17 marks)

a) PaPa added 10% GST onto the price of a book valued at \$29.90 to get its retail price. He then discounted the retail price, by 10% to get the 'on sale' price.

(i) What was the retail price? **1**

(ii) What is the 'on sale' price? **1**

(iii) By what percentage should PaPa have discounted the retail price to get back to what the book was valued at? **2**

Question 2 Continued

b) Simplify the following expressions, leaving each with only positive indices:

(i) $x^5 \times x^7$ **1**

(ii) $p^{-\frac{1}{2}} \times p^{\frac{1}{3}}$ **2**

(iii) $\dfrac{6a^5b^6}{4a^3b^2}$ **1**

(iv) $\dfrac{2x^2+6}{8x^6+24x^4}$ **2**

Question 2 Continued

c) A line goes through the points $A(1, 3)$ and $B(-3, 4)$.

(i) Find the midpoint of the interval joining A and B. **1**

(ii) Find the gradient of the straight line through A and B. **1**

(iii) Find the exact distance from A to B. **1**

(iv) What is the equation of the straight line passing through both A and B? Leave your answer in general form. **2**

(v) Find the equation of the line passing through $(5, 7)$ and perpendicular to the line passing through A and B. **2**

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Question 3 (16 marks)

a) Simplify $\left(\sqrt{5}-3\right)\left(\sqrt{5}+3\right)$. **1**

b) Solve the following inequality and show the solution on a number line. **2**

$17-2m\leq 16$

c) Make x the subject of the formula below, leaving your answer as a simplified fraction. **2**

$$\frac{x+y}{y}=\frac{y}{a}+\frac{a}{y}$$

Question 3 Continued

d) Solve for x. **2**

$$\frac{4x-1}{3}-\frac{2x+3}{4}=\frac{x+1}{6}$$

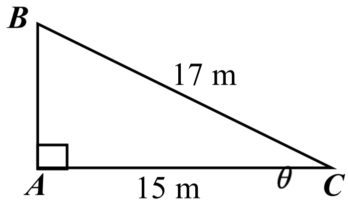
e) Fully factorise the following expressions:

(i) $100x^2-81y^2$ **1**

(ii) $x^2-9x-36$ **1**

Question 3 Continued

f) With respect to the diagram below, write down the value of:

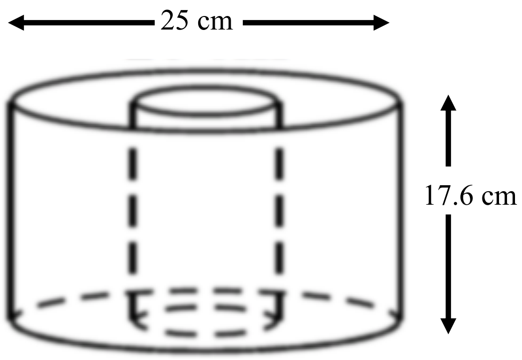


(i) $\cos\theta$ **1**

(ii) $\tan\theta$ **2**

(iii) θ (nearest minute) **1**

g) Find the surface area of this solid correct to one decimal place. **3**



Diameter of hole 8 cm

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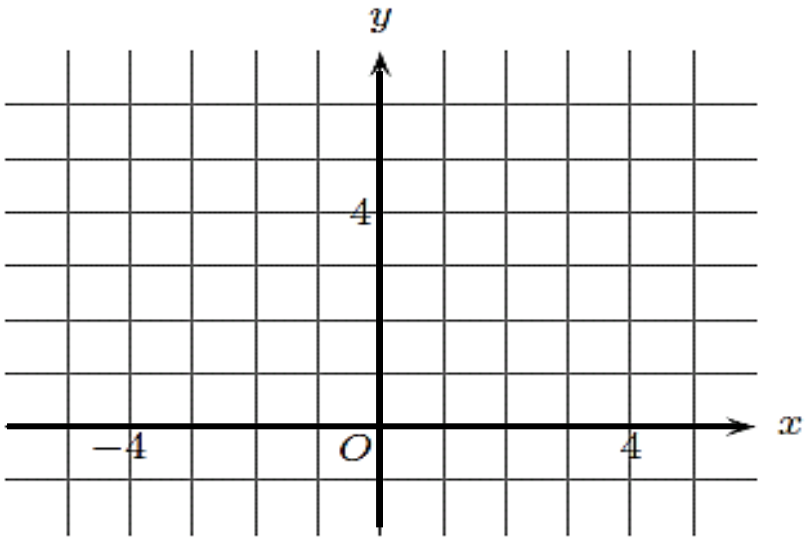
Question 4 (16 marks)

a) By using graphical means, find the point of intersection of $x+3y=1$ and $y=2x+5$. **3**



b) Fully factorise the following expressions:

(i) $12x^2-23x+5$ **2**

(ii) $x^3-25x-3x^2+75$ **2**

Question 4 Continued

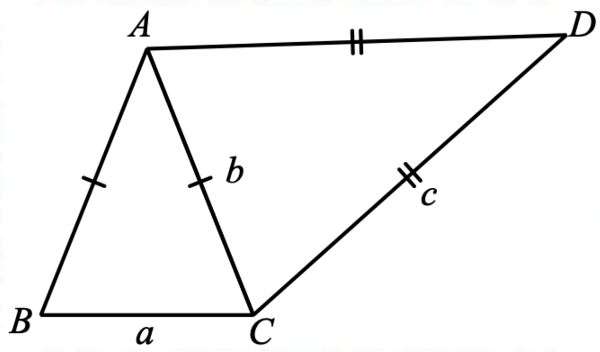
c) Express $\dfrac{2\sqrt{3}}{\sqrt{3}-1}$ with a rational denominator. **2**

d) A packet of lollies contains 5 red lollies and 15 green lollies. Two lollies are selected at random without replacement. By drawing a tree diagram, or otherwise, find the probability that the two lollies are different colours. **3**

Question 4 Continued

e) In the diagram below, both ΔABC and ΔDAC are isosceles triangles.

$AB = AC$, $AD = CD$ and $\angle BAC = \angle ADC$.



NOT TO SCALE

(i) Prove that $\Delta ABC \;|||\; \Delta DAC$. **2**

(ii) Hence prove that $b^2 = ac$. **2**

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Question 5 (17 marks)

a) Consider the 2019-2020 yearly tax table below.

Mrs West earns \$42 042 dollars per year, and \$567 income from other sources.
She has deductions of \$665, and has paid PAYE (Pay As You Earn) tax instalments totalling \$4042.

With the help of the table, find:

Taxable income	Tax on this income
0 – \$18,200	Nil
\$18,201 – \$37,000	19c for each \$1 over \$18,200
\$37,001 – \$90,000	\$3,572 plus 32.5c for each \$1 over \$37,000
\$90,001 – \$180,000	\$20,797 plus 37c for each \$1 over \$90,000
\$180,001 and over	\$54,097 plus 45c for each \$1 over \$180,000

The above rates **do not** include the Medicare levy of 2%.

(i) her taxable income. **1**

(ii) the tax payable on her taxable income. **2**

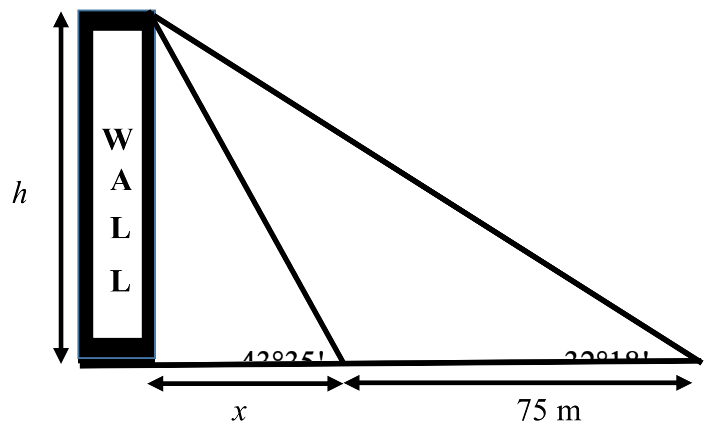
(iii) the refund, or balance payable, after the Medicare levy of 2% is paid. **2**

b) Express as a single fraction, with a rational denominator: **2**

$$\sin(60°)+\tan(30°)+\cos(0°)$$

<u>Question 5 Continued</u>

c) Consider the diagram below, showing a wall supported by two stays.



(i) Find two expressions for the height of the wall h, in terms of x. **2**

(ii) Solve for x. **1**

(iii) Find the height of the wall. **1**

Question 5 Continued

d) Simplify: **3**

$$\frac{x^2-25}{3x^2+15x}-\frac{x^2+4x-5}{x^2-x}$$

e) Over 1 year you want to invest \$30 000. **3**
Part of this will be invested in a 5% p.a. simple interest rate account.
The remainder will be invested in the family company returning 7% p.a (simple interest) on the investment.
What is the least you can invest in the family company and still get at least \$1900 interest?

(Hint: construct an inequality where x is the amount invested in the family company.)

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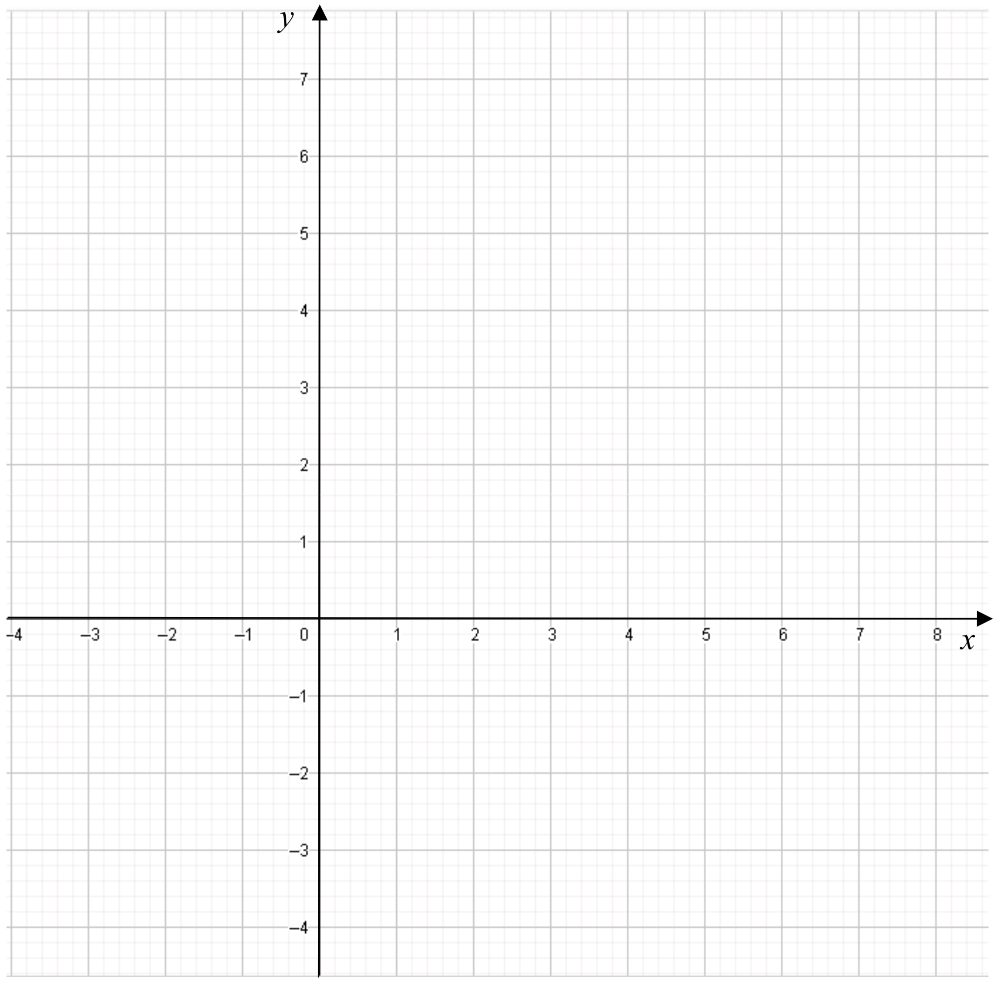
Question 6 (18 marks)

a) On the axes below graph and label the following lines:

(i) $y = -1$ **1**

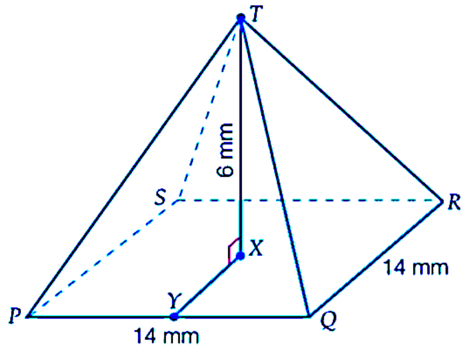
(ii) $y = \dfrac{2x}{3}$ **1**

(iii) $y + 2x = 6$ **2**



Question 6 Continued

b)



In the diagram above $PQRST$ is a right pyramid, Y is the midpoint of PQ and X is the centre of the square base $PQRS$.

(i) Show that $\angle TYX = 40°36'$ (nearest minute). **1**

(ii) Show that $\angle TQX = 31°13'$ (nearest minute). **3**

c) Which number's square root is equal to two times its cube root? **2**

d) A game involves rolling two six-sided dice, followed by rolling a third six-sided die. To win the game, the number rolled on the third die must lie between the two numbers rolled previously. For example, if the first two dice show 1 and 4, the game can only be won by rolling a 2 or 3 with the third die.

(i) What is the probability that a player has no chance of winning before rolling the third die? **2**

(ii) What is the probability that the player wins the game? **2**

Question 6 Continued

e) A sequence is *arithmetic* if the difference between any two consecutive terms is always the same i.e. constant.

This means that if the *n* numbers $a_1,\ a_2,\ a_3,\ \ldots,\ a_{n-1},\ a_n$ form an arithmetic sequence then $a_2 - a_1 = a_3 - a_2 = \ldots = a_n - a_{n-1} = d$, for some constant *d*.

(i) Find an expression for $a_n - a_1$ in terms of *n* and *d*. **1**

(ii) If $\frac{1}{x_1},\ \frac{1}{x_2},\ \frac{1}{x_3},\ \ldots,\ \frac{1}{x_n}$ form an arithmetic sequence, show that **3**

$$x_1x_2 + x_2x_3 + x_3x_4 + \ldots + x_{n-1}x_n = (n-1)x_1x_n.$$

End of paper

SYDNEY
BOYS
HIGH
SCHOOL

2019

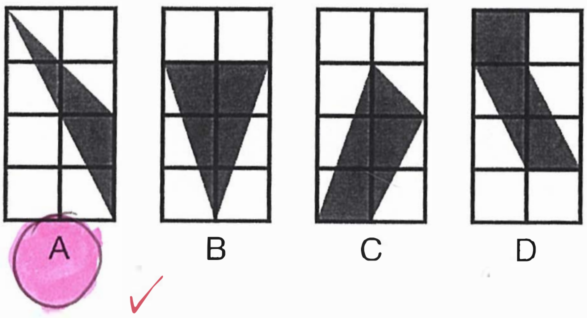
YEAR 9 YEARLY
ASSESSMENT TASK

Mathematics Stage 5.3

SUGGESTED SOLUTIONS

Question 1 (16 marks)

a) Circle the letter of the polygon that does not have the same area as the others. **1**



A: $\frac{1}{2} \times 2 \times 2 = 2$

B: $\frac{1}{2} \times 2 \times 3 = 3$

C: $(\frac{1}{2} \times 1 \times 3) \times 2 = 3$

D: $1^2 + 1 \times 2 = 3$

✓

b) Expand then collect like terms for the following expressions:

(i) $4(a+1)-6a-5$ **1**

Some minor mistakes with expansion

$4a + 4 - 6a - 5$

$= -2a - 1$ ✓

(ii) $(2a+b)(a-11b)$ **1**

$2a^2 - 22ab + ab - 11b^2$

$= 2a^2 - 21ab - 11b^2$ ✓

c) JoJay, a casual labourer, is paid \$21.78 per hour. How much will he earn if he works a 35 hour week? **1**

2dp for \$

$35 \times \$21.78$

$= \$762.30$ ✓

$-\frac{1}{2}$ if not answered in 2dp.

d) Show by substitution that $a = 2$ is the solution to the following equation. **1**

$11(a+8) = 5(a+20)$

By substitutio

$-\frac{1}{2}$ if student solve the eq

Some students neglected to read "by substitution"

$11(10) = 5(22)$ ✓

$110 = 110$

$\therefore a = 2$ is a solution

e) Write $2500 \times 3\,640\,000$ in scientific notation. **1**

Some students read this as "rewrite the question in sci. notation." No $\frac{1}{2}$ mark

9100000000

9.1×10^9 ✓

Question 1 Continued

f) Solve the following equations, giving answers in simplest exact form.

(i) $4p+6=37$ **1**

$4p=31$

$p=\frac{31}{4}$ or 7.75 ✓

(ii) $\frac{y}{5}+3=7$ **1**

$\frac{y}{5}=4$

$y=20$ ✓

(iii) $2y^2+9=23$ **2**

$2y^2=14$

$y^2=7$

$y=\pm\sqrt{7}$ ✓✓

Most students did not realise $(-\sqrt{7})^2=+7$

$-\frac{1}{2}$ for not having $-\sqrt{7}$.

(iv) $\frac{2x-1}{4}=\frac{3x+2}{8}$ **2**

$8(2x-1)=4(3x+2)$

$16x-8=12x+8$

$4x=16$ ✓

$x=4$ ✓

Question 1 Continued

g) Express $5x^{-3}$ with a positive index. **1**

$\frac{5}{x^3}$ ✓ $\frac{1}{2}$ for $\frac{1}{5x^3}$

h) Find the gradient m and y-intercept of these lines:

(i) $y = 3x + 2$ **1**

$m = 3$ ✓ $\frac{1}{2}$

y-intercept $= 2$ $\frac{1}{2}$

1

(

(ii) $4x + 3y + 6 = 0$ **2**

$3y = -4x - 6$

$y = -\frac{4}{3}x - 2$ ✓

$m = -\frac{4}{3}$ $\frac{1}{2}$ ✓

y-intercept $= -2$. $\frac{1}{2}$

Question 2 (17 marks)

a) PaPa added 10% GST onto the price of a book valued at $29.90 to get its retail price. He then discounted the retail price, by 10% to get the 'on sale' price.

(i) What was the retail price? **1**

$$\begin{array}{r} 29.90 + \\ 2.99 \\ \hline \$32.89 \end{array}$$

(ii) What is the 'on sale' price? **1**

$0.9 \times \$32.89$

$= 29.601$

$= \$29.60$

(iii) By what percentage should PaPa have discounted the retail price to get back to what the book was valued at? **2**

$\$32.89 \times x = \$29.90$

$x = \dfrac{29.90}{32.89}$

$x = 90.9\%$

$\therefore \%\ \text{discount} = 9.1\%$

2

Question 2 Continued

b) Simplify the following expressions, leaving each with only positive indices:

(i) $x^5 \times x^7$ **1**

x^{12}

(ii) $p^{-\frac{1}{2}} \times p^{\frac{1}{3}}$ **2**

$p^{-\frac{3}{6}+\frac{2}{6}} = p^{-\frac{1}{6}} = \frac{1}{p^{\frac{1}{6}}}$

2

or $\frac{1}{\sqrt[6]{p}}$

(iii) $\frac{\cancel{6}^{3}a^5b^6}{\cancel{4}_{2}a^3b^2} = \frac{3a^2b^4}{2}$ **1**

(iv) $\frac{2x^2+6}{8x^6+24x^4} = \frac{\cancel{2}^{1}\cancel{(x^2+3)}}{4\cancel{8}x^4\cancel{(x^2+3)}}$ **2**

2

$= \frac{1}{4x^4}$

Question 2 Continued

c) A line goes through the points $A(1, 3)$ and $B(-3, 4)$.

(i) Find the midpoint of the interval joining A and B. **1**

$M = \left(\frac{1+-3}{2}, \frac{3+4}{2}\right) = (-1, 2)$ ✓

(ii) Find the gradient of the straight line through A and B. **1**

$m_{AB} = \frac{4-3}{-3-1} = \frac{1}{}$ $-\frac{1}{4}$ ✓

(iii) Find the exact distance from A to B. **1**

$d = \sqrt{(1--3)^2 + (3-4)^2}$

$d = \sqrt{16 + 1} = \sqrt{17}$ ✓

(iv) What is the equation of the straight line passing through both A and B? Leave your answer in general form. **2**

$y - y_1 = \quad (x - \quad)$

$y - 3 = -\frac{1}{4}(x-1)$ ✓

$4y - 12 = -x + 1$ ✓

$x + 4y - 13 = 0$ ✓

(2)

(v) Find the equation of the line passing through (5, 7) and perpendicular to the line passing through A and B. **2**

line $\perp$ to AB has $m = 4$ ✓

Eqn is $y - 7 = 4(x-5)$

$y - 7 = 4x - 20$

$y = 4x - 13$ ✓

(2

Question 3 (16 marks)

a) Simplify $\left(\sqrt{5}-3\right)\left(\sqrt{5}+3\right)$. **1**

$= 5-9 = -4$

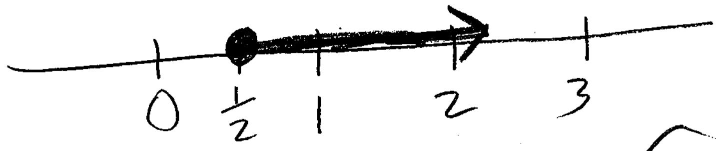
1

b) Solve the following inequality and show the solution on a number line. **2**

$17-2m\le 16$

$-2m \le -1$

$m \ge \frac{1}{2}$



2

c) Make x the subject of the formula below, leaving your answer as a simplified fraction. **2**

$$\frac{x+y}{y}=\frac{y}{a}+\frac{a}{y}$$

$\times ay \Rightarrow a(x+y) = y^2+a^2$

$x+y = \frac{a^2+y^2}{a}$

$x = \frac{a^2+y^2}{a} - y$

$x = \frac{a^2+y^2-ay}{a}$

2

Question 3 Continued

d) Solve for x. **2**

$$\frac{4x-1}{3}-\frac{2x+3}{4}=\frac{x+1}{6}$$

$\times 12 \Rightarrow 4(4x-1)-3(2x+3)=2(x+1)$

$16x-4-6x-9=2x+2$

$10x-2x=2+13$

$8x=15$

$x=\frac{15}{8}$

2

e) Fully factorise the following expressions:

(i) $100x^2-81y^2$ **1**

$=(10x)^2-(9y)^2$

$=(10x-9y)(10x+9y)$

1

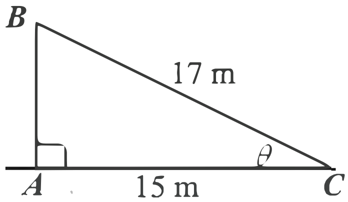
(ii) $x^2-9x-36$ **1**

$=(x-12)(x+3)$

1

Question 3 Continued

f) With respect to the diagram below, write down the value of:



(i) $\cos\theta$ **1**

$= \frac{15}{17}$ 1

(ii) $\tan\theta$ **2**

$AB = \sqrt{17^2 - 15^2} = \sqrt{64} = 8$ 2

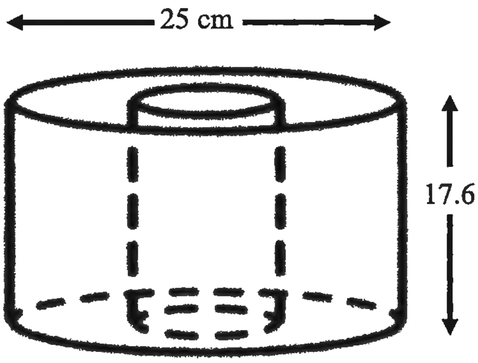
$\tan\theta = \frac{8}{15}$

(iii) θ (nearest minute) **1**

$28°4'$ 1

g) Find the surface area of this solid correct to one decimal place. **3**

Let $R = 12.5$, $r = 4$



Diameter of hole 8 cm

$SA = 2\pi R^2 + 2\pi Rh - 2\pi r^2 + 2\pi rh$

$= 2\pi(12.5)^2 + (2\pi \times 12.5 \times 17.6) - 2\pi(16) + (2\pi \times 4 \times 17.6)$

$= 312.5\pi + 440\pi - 32\pi + 140.8\pi$

$= 752.5\pi - 32\pi + 140.8\pi$

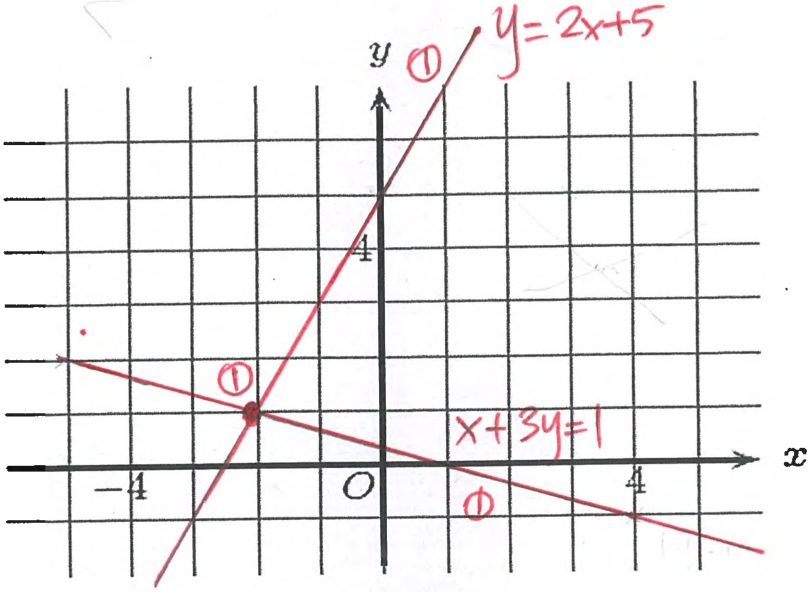
$= 981.75 + 1382.30 - 100.5 + 442.37$

$SA = 2705.856$

$SA = 2705.9\ cm^2$ 3

Question 4 (16 marks)

a) By using graphical means, find the point of intersection of $x+3y=1$ and $y=2x+5$. **3**



∴ the point of intersection is $(-2, 1)$

b) Fully factorise the following expressions:

(i) $12x^2-23x+5$ **2**

$(3x-5)(4x-1)$ ① ①

OR

$(-3x+5)(-4x+1)$ ① ①

$$\begin{array}{c} 3x \quad -5 \\ 4x \quad -1 \\ \hline -3x-20x \end{array}$$

$$\begin{array}{c} -3x \quad 5 \\ -4x \quad 1 \\ \hline -3x-20x \end{array}$$

* incorrect signs (-0.5)

(ii) $x^3-25x-3x^2+75$ **2**

$= x^3-25x-3(x^2-25)$

$= x(x^2-25)-3(x^2-25)$ ①

$= (x-3)(x^2-25)$ (0.5)

$= (x-3)(x+5)(x-5)$ (0.5)

Question 4 Continued

c) Express $\dfrac{2\sqrt{3}}{\sqrt{3}-1}$ with a rational denominator. **2**

$$\frac{2\sqrt{3}\,\boxed{(\sqrt{3}+1)}}{(\sqrt{3}-1)\,\boxed{(\sqrt{3}+1)}} \quad 0.5$$

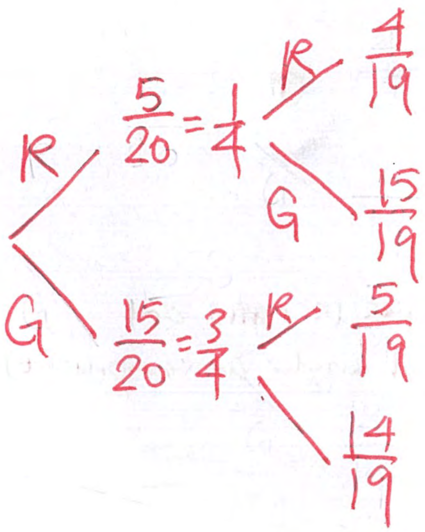
$$= \frac{6+2\sqrt{3}}{3-1}$$

$$= \frac{6+2\sqrt{3}}{2} \quad 1$$

$$\frac{2(3+\sqrt{3})}{2}$$

$$= 3+\sqrt{3} \quad 0.5$$

d) A packet of lollies contains 5 red lollies and 15 green lollies. Two lollies are selected at random without replacement. By drawing a tree diagram, or otherwise, find the probability that the two lollies are different colours. **3**



tree diagram 1

P(RG, GR)

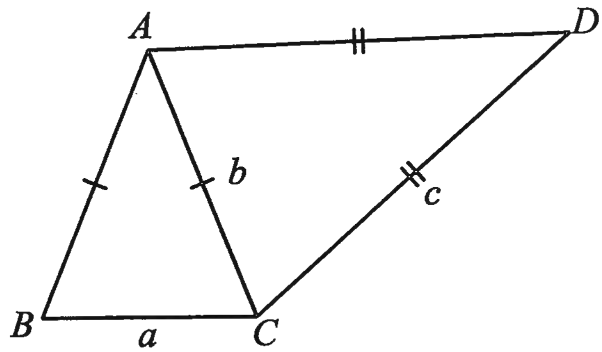
$$= \boxed{\frac{1}{4}\times\frac{15}{19}} + \boxed{\frac{3}{4}\times\frac{5}{19}} \quad 1 \quad 1$$

$$= \frac{15}{76}+\frac{15}{76} = \frac{30}{76} = \frac{15}{38}$$

Question 4 Continued

e) In the diagram below, both ΔABC and ΔDAC are isosceles triangles.

$AB = BC$, $AD = CD$ and $\angle BAC = \angle ADC$.



NOT TO SCALE

(i) Prove that $\Delta ABC \parallel\!\mid \Delta DAC$. **2**

$\because \angle BAC = \angle ADC$ (given) (0.5)

$AB = AC$ (given)

$AD = DC$ (given)

$\therefore \frac{AB}{AD} = \frac{AC}{DC}$ (1)

$\therefore \Delta ABC \parallel\!\mid \Delta DAC$ (two sides in ratio and included angles are congruent) (0.5)

(ii) Hence prove that $b^2 = ac$. **2**

$\because \Delta ABC \parallel\!\mid \Delta DAC$ (proved)

I II

$\therefore \frac{BC}{AC} = \frac{AC}{DC}$ (corresponding sides of similar Δs have the same ratio) (1)

* if only II + III (1.5)

III

$\frac{a}{b} = \frac{b}{c}$ (0.5)

$b^2 = ac$

Solutions for question 5 of the Year 9 Yearly Exam, 2019

a) i) Mrs West's total income included her wages and income from other sources. The deductions must come off her taxable income.
$42 042 + $567 - $665 = $41 944

1 mark for correct solution and half a mark deducted for incorrectly adding or subtracting the right amount or an amount that isn't involved in the solution.

ii) Tax payable is:
($41 944 - $37 000) $\times$ 0.325 + $3 572 = $5 178.80

Half marks awarded for each appropriate operation.

iii) The medicare levy is based on taxable income. It is 2% of $41 944, which is equal to $838.88.
Mrs West has already payed the ATO $4 042.
Amount payable is:
$5 178.80 + $838.88 - $4 042
= $1 975.68

1 mark for getting the correct value for the levy. Half mark awarded for adding it to part i) and another half mark for subtracting the PAYG.

b) $\sin(60°) + \tan(30°) + \cos(0°)$

$= \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{3}} + 1$

$= \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{3} + 1$

$= \frac{3\sqrt{3}}{6} + \frac{2\sqrt{3}}{6} + \frac{6}{6}$

$= \frac{6+5\sqrt{3}}{6}$

1 mark for using all the correct exact values for the ratios. 1 mark for getting the correct answer (half marks were deducted for single mistakes in either of these two areas).

c) i) $\tan(43°35') = \frac{h}{x} \rightarrow h = x \times \tan(43°35')$

similarly: $h = (x + 75) \times \tan(32°18')$

1 mark was awarded for each expression. Only half a mark was awarded to an expression for x in terms of h. No marks if your expression involved more than 2 unknowns.

ii) $x \times \tan(43°35') = (x + 75) \times \tan(32°18')$

$$\frac{\tan(43°35')}{\tan(32°18')} = \frac{x+75}{x}$$

$$RHS = 1 + \frac{75}{x}$$

$$\rightarrow \frac{\tan(43°35')}{\tan(32°18')} - 1 = \frac{75}{x}$$

$$\rightarrow x = \frac{75}{\frac{\tan(43°35')}{\tan(32°18')} - 1}$$

$$\rightarrow x = \frac{75 \times \tan(32°18')}{\tan(43°35') - \tan(32°18')}$$

$x = 148.37$ m to the nearest centimeter

Half a mark awarded for equating the expressions in i) and another half for the correct answer.

iii) substitute x into either expression in part i) to find h.

$h = 148.37 \times \tan(43°35')$

$h = 141.21$ to the nearest centimeter.

Half a mark awarded for putting the answer for ii) into the appropriate expression and half a mark awarded for the answer.

d)

$$\frac{x^2 - 25}{3x^2 + 15x} - \frac{x^2 + 4x - 5}{x^2 - x}$$

$$\rightarrow \frac{(x-5)(x+5)}{3x(x+5)} - \frac{(x+5)(x-1)}{x(x-1)}$$

$$\rightarrow \frac{x-5}{3x} - \frac{x+5}{x}$$

$$\rightarrow \frac{x-5-3(x+5)}{3x}$$

$$\rightarrow \frac{x-5-3x-15}{3x}$$

$$\rightarrow -\frac{2(x+10)}{3x}$$

1 mark for factorising each numerator and denominator. Half a mark awarded for correctly equating similar factors in the numerator and denominator to 1 in each term. 1 mark for correctly cross multiplying and half a mark for the correct answer. Half marks deducted for each simple error.

e) Let x be the portion (or fraction) of the \$30 000 to be invested into the family company.

Then the amount of interest from the investment is $x \times \$30{,}000 \times 7\%$.

The rest of the investment can be described in terms of x also. $1 - x$ is the portion of the \$30 000 that will be invested into a simple interest account.

Then the amount of interest from the investment is $(1 - x) \times \$30{,}000 \times 5\%$.

The sum of these must be at least \$1900.

$$\$1900 = (1 - x) \times \$30{,}000 \times 5\% + x \times \$30{,}000 \times 7\%$$

$$\$1900 = \$30\ 000 \times 5\% + x \times \$30{,}000(7\% - 5\%)$$

$$x \times \$30{,}000 \times 2\% = \$1900 - \$30{,}000 \times 5\%$$

$$x = \frac{\$1900 - \$30{,}000 \times 5\%}{\$30{,}000 \times 2\%}$$

$$x = \frac{1900 - 1500}{600}$$

$$x = \frac{400}{600} \to \frac{2}{3}$$

Therefore the minimum amount that needs to be invested into the family company to earn \$1900 per year in interest is two thirds of \$30 000, i.e. \$20 000.

Bald answers only received 1 mark. 1 mark for writing a correct expression that would result in the correct answer. 1 mark for making x the subject and 1 mark for the correct answer (i.e. x is the least amount of money needed to be invested into the family company).

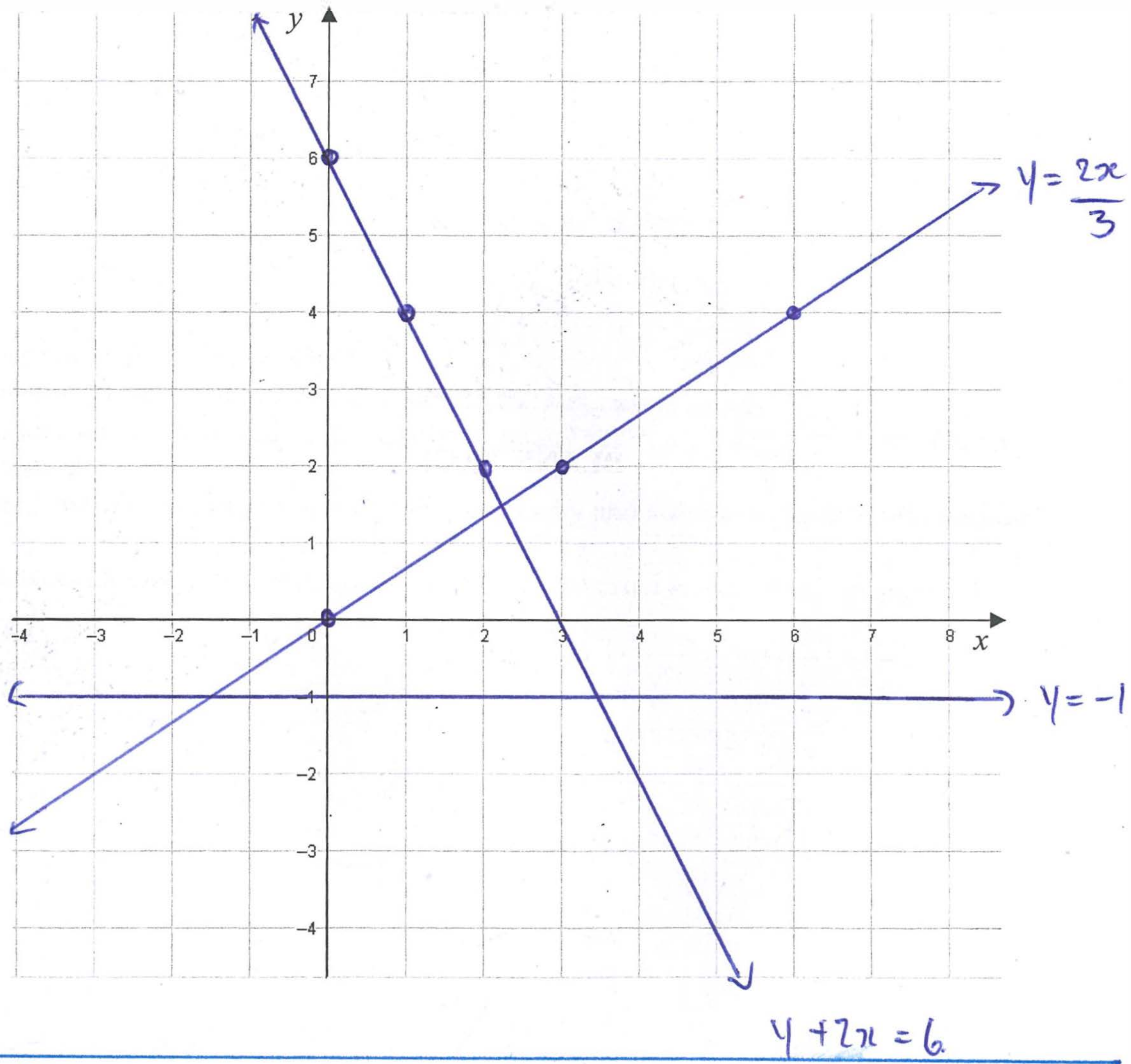
Question 6 (18 marks)

a) On the axes below graph and label the following lines:

(i) $y = -1$ **1**

(ii) $y = \dfrac{2x}{3}$ **1**

(iii) $y + 2x = 6$ $y = -2x + 6$ **2**



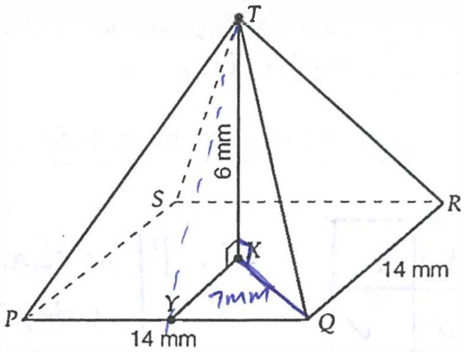
Marker's Comments

(1) No half marks for (i). If candidates drew a vertical line at $x = -1$, no marks awarded.

(2) For (iii), 1 mark for correct graph and 1 mark for both intercepts.

Question 6 Continued

b)



In the diagram above *PQRST* is a right pyramid, *Y* is the midpoint of *PQ* and *X* is the centre of the square base *PQRS*.

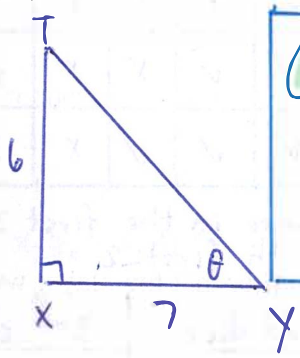
(i) Show that $\angle TYX = 40°36'$ (nearest minute). **1**

let $\angle TYX = \theta$

$\tan\theta = \frac{6}{7}$

$\theta = \tan^{-1}\left(\frac{6}{7}\right)$

$\approx 40°36'$



1 mark for expressing $\theta = \tan^{-1}\left(\frac{6}{7}\right)$

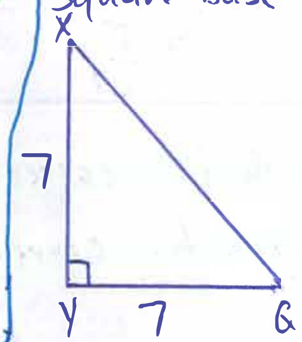
(ii) Show that $\angle TQX = 31°13'$ (nearest minute). **3**

1 mark for using Pythagoras' theorem

1 mark for finding $\sqrt{98}$

1 mark for $\angle TQX = 31°13'$

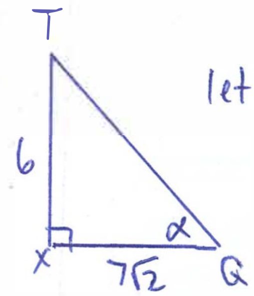
looking at the top diagram at the square base



By Pythagoras' theorem

$XQ = \sqrt{7^2+7^2}$

$= 7\sqrt{2}$



let $\angle TQX = \alpha$

$\tan\alpha = \frac{6}{7\sqrt{2}}$

$\alpha = \tan^{-1}\left(\frac{6}{7\sqrt{2}}\right)$

$\approx 31°13'$

c) Which number's square root is equal to two times its cube root? **2**

let the number be x

$\sqrt{x} = 2\sqrt[3]{x}$

$\frac{x^{1/2}}{x^{1/3}} = 2$

$x^{1/6} = 2$

$x = 2^6 = 64$

~~AN~~

Also $x = 0$ is also solution.

1 mark for $x = 64$

1 mark for $x = 0$

Question 6 Continued

d) A game involves rolling two six-sided dice, followed by rolling a third six-sided die. To win the game, the number rolled on the third die must lie between the two numbers rolled previously. For example, if the first two dice show 1 and 4, the game can only be won by rolling a 2 or 3 with the third die.

(i) What is the probability that a player has no chance of winning before rolling the third die? **2**

Die 1 \ Die 2	1	2	3	4	5	6
1	X	X	✓	✓	✓	✓
2	X	X	X	✓	✓	✓
3	✓	X	X	X	✓	✓
4	✓	✓	X	X	X	✓
5	✓	✓	✓	X	X	X
6	✓	✓	✓	✓	X	X

X No chance in the first 2 die

✓ Yes with first 2

$\therefore P\left(\text{no chance winning with first 2}\right) = \frac{16}{36} = \frac{4}{9}$

* 1 mark: For working towards a correct answer.
* 1 mark: Correct answer.
* 0 mark: If just a wrong answer with no working.

(ii) What is the probability that the player wins the game? **2**

First dice	2nd dice	3rd dice
1	3	2
3	1	2
1	4	2, 3
4	1	3, 2
1	5	2, 3, 4
5	1	2, 3, 4
1	6	2, 3, 4, 5
6	1	2, 3, 4, 5
2	4	3
4	2	3
2	5	3, 4
5	2	3, 4
2	6	3, 4, 5
6	2	3, 4, 5
3	5	4
5	3	4
3	6	4, 5
6	3	4, 5
4	6	5
6	4	5

$P(\text{win}) = \frac{40}{6\times6\times6}$

$= \frac{40}{216}$

$= \frac{5}{27}$

(1) mark for correct working

(1) mark for correct solution

Question 6 Continued

e) A sequence is *arithmetic* if the difference between any two consecutive terms is always the same i.e. constant.

This means that if the n numbers a_1, a_2, a_3, …, a_{n-1}, a_n form an arithmetic sequence then $a_2 - a_1 = a_3 - a_2 = ... = a_n - a_{n-1} = d$, for some constant d.

(i) Find an expression for $a_n - a_1$ in terms of n and d. **1**

$a_2 - a_1 = d$

$a_3 - a_1 = 2d$

$a_4 - a_1 = 3d$

⋮

$\boxed{a_n - a_1 = (n-1)d}$

1 mark for correct solution

(ii) If $\frac{1}{x_1}, \frac{1}{x_2}, \frac{1}{x_3}, \ldots, \frac{1}{x_n}$ form an arithmetic sequence, show that **3**

$$x_1x_2 + x_2x_3 + x_3x_4 + ... + x_{n-1}x_n = (n-1)x_1x_n.$$

Since $\frac{1}{x_1}, \frac{1}{x_2}, \frac{1}{x_3}, \cdots \frac{1}{x_n}$ is an arithmetic series

$\frac{1}{x_2} - \frac{1}{x_1} = d$

$\therefore \frac{x_1 - x_2}{x_1 x_2} = d \quad \therefore x_1x_2 = \frac{x_1 - x_2}{d}$

$\frac{1}{x_3} - \frac{1}{x_2} = d$

$\frac{x_2 - x_3}{x_2 x_3} = d$

$\therefore x_2x_3 = \frac{x_2 - x_3}{d}$

$\frac{1}{x_n} - \frac{1}{x_{n-1}} = d$

$\frac{x_{n-1} - x_n}{d} = x_n x_{n-1}$

$\therefore LHS = x_1x_2 + x_2x_3 + \ldots + x_{n-1}x_n$

$= \frac{x_1 - x_2}{d} + \frac{x_2 - x_3}{d} + \frac{x_3 - x_4}{d} + \cdots + \frac{x_{n-1} - x_n}{d}$

$= \frac{x_1 - x_n}{d}$ ①

From (i)

$\frac{1}{x_n} - \frac{1}{x_1} = (n-1)d$

$\frac{x_1 - x_n}{x_n x_1} = (n-1)d$

$\frac{x_1 - x_n}{d} = (n-1)x_n x_1$ ②

Sub ① into ②

$\therefore LHS = (n-1)x_n x_1$

$= RHS.$

1 mark for finding an expression for $x_n x_{n-1}$

1 mark for finding simplified expression for LHS.

1 mark to show LHS = RHS

End of paper